

Future Tech: The Next Generation

All facets of the human life improve with each scientific innovation, here are some of those innovations

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"Any sufficiently advanced technology is indistinguishable from magic."

– Arthur C. Clarke

The challenges humanity has overcome with innovations in the sciences and technology is staggering; from the first use of stones to butcher dead animals by the Olduvai in Tanzania nearly 2.5 million years ago, to the upcoming use of electromagnetic thrusters in the Variable Specific Impulse Magnetoplasma Rocket to travel to Mars at speeds of up to 1,80,000 kmph. In the meanwhile, we as a species have split the atom, put a man on the moon, rewrote the genome, connected humanity and developed technology that can store the archive of human creation in our pockets. And we're not done yet.

The following are some of the most critically relevant technologies of the coming years that will not only shape the way we live our lives but also determine how well we live it.

Grid-scale Electricity Storage

The problem with power has always been that of storage. We could develop the most revolutionary source of renewable energy in the future but without an effective means of storage most of it would go to waste. This is where research into grid scale electricity storage comes into play. The next generation of innovation is aimed towards fixing this gap and



A battery made using everyday materials being assembled at Urban Electric

make possible efficient storage of massive amounts of energy for future use.

So far numerous different methods of energy storage have been developed, such as flywheels, compressed air, batteries, hydrogen storage, pumped-water storage (dams), superconducting magnetic energy storage and thermal storage. Each of these methods have varying efficiencies, for example – batteries allow for at the very least 50 percent of the energy stored to be retrieved whereas thermal storage techniques can potentially allow 99 percent

energy retrieval. But these efficiencies have all been pushed to the maximum range in different projects around the world and broadly range between 90 to 99 percent efficient storage. The real challenge is making them cost effective.

In this regard most of the investment has been in large scale batteries; a key example is the largest grid scale battery ever created, in Sendai Miyagi Japan which has a storage capacity of 40,000 kilowatts (40 megawatts) and is expected to be online in early 2015. This pilot project by Toshiba will augment the existing power grid with energy generated from the variable wind and solar generators. On the other side of the world, chemical engineer Sanjoy Banerjee's start-up company in Harlem, New York called Urban Electric has developed a cheap, safe and efficient battery in a miniaturised form using everyday non-toxic materials.

Alternatives to battery technology which are better linked to renewable



Sendai Substation Li-ion battery pilot project